

Notes: Prescott and Visscher (JPE, 1980)

Organization Capital

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Model objects and measurement	3
3.1	Organization capital	3
3.2	Personnel information	3
3.3	Task structure	3
3.4	Signals and Bayesian learning	3
3.5	Growth rate and screening time	4
3.6	Team information	4
3.7	Firm-specific human capital	4
4	Models and theoretical specifications	4
4.1	Cost function in the personnel-information model	4
4.2	Growth raises per-unit costs	4
4.3	Technology constraints and constant returns	5
4.4	Industry equilibrium and firm growth	5
4.5	Transferability and firm specificity	5
5	Identification and theoretical design logic	5
5.1	What the paper explains	5
5.2	Main assumptions that matter	6
5.3	Limitations	6
6	Section-by-section study guide	6
6.1	Introduction	6
6.2	Section I: Personnel Information as Organization Capital	6
6.3	Section II: Team Information	6
6.4	Section III: Firm-specific Human Capital	6
6.5	Section IV: Costs of Transferring Organization Capital	6
6.6	Section V: Summary and Concluding Remarks	7
7	Tables, figures, and theoretical result blocks	7
7.1	Result Block 1: Organization Capital as Information	7
7.2	Result Block 2: Personnel Information and Screening	7
7.3	Result Block 3: Endogenous Adjustment Costs	8

7.4	Result Block 4: Constant Returns and Gibrat's Law	8
7.5	Result Block 5: Team Information	8
7.6	Result Block 6: Firm-specific Human Capital	9
7.7	Result Block 7: Transfer Costs and Firm Boundaries	9
8	Detailed result map	9
9	Short summary	10
10	Conclusion	10
10.1	Core conclusion	10
10.2	Economic intuition	10
10.3	Incremental contribution revisited	10
10.4	Implications for the theory of the firm	11
10.5	Implications for firm growth and size distributions	11
10.6	Limitations	11
10.7	Takeaway	11

1 Big picture

- **Question.** What is a firm, and why do firms exhibit gradual growth rather than costless instantaneous expansion when profitable opportunities appear?
- **Core answer.** A firm is a stock of information. Prescott and Visscher call this stock **organization capital**: information about employees, tasks, teams, and firm-specific human capital that affects the production possibility set.
- **Main mechanism.** Organization capital is produced jointly with output. Growing faster forces the firm to use workers, teams, or human capital before enough information has accumulated. This raises production costs and creates endogenous adjustment costs.
- **Why the firm exists.** The firm is an institution that facilitates the accumulation, transmission, and use of information in a less costly way than many alternative market arrangements.
- **Why growth is gradual.** In each model, current production competes with the accumulation of organization capital. Rapid growth lowers the average amount of information per worker/team/task and therefore raises costs.
- **Why Gibrat's Law appears.** The technology set is a convex cone with constant returns to scale and internal adjustment costs. This produces an optimal growth rate that is independent of firm size.
- **Why firms do not instantly merge or pirate all productive employees.** Organization capital is partly firm specific and costly to transfer. Without transfer costs, the model would not constrain firm growth.
- **Broad contribution.** The paper links the theory of the firm, learning, adjustment costs, firm growth, and size distributions through a unified information-capital mechanism.

2 Incremental contribution of the paper

- **A theory of organization capital.** The paper formalizes the idea that firm-specific information is a capital stock. This moves beyond treating the firm only as a production function or a contract nexus.
- **Endogenous adjustment costs.** Rather than assuming convex adjustment costs, the paper derives them from information accumulation: rapid expansion requires less informed assignments and weaker organizational learning.
- **Information-based theory of firm growth.** The model explains gradual firm growth and Gibrat's Law using organizational information rather than managerial span of control, financing constraints, or technological indivisibilities.
- **Connection to the nature of the firm.** The paper offers an information-processing interpretation of why firms exist: they create incentives and structures to acquire and transmit information internally.
- **Multiple forms of organization capital.** The paper studies personnel information, team information, and firm-specific human capital as distinct but mathematically related forms of organization capital.
- **Constant returns with internal frictions.** The paper reconciles two facts: firms can have approximately constant returns to scale, yet still face costly rapid expansion.
- **Firm-size distribution implications.** By generating size-independent optimal growth rates, the model provides a microfoundation for Gibrat-style firm growth and skewed firm-size distributions.
- **Early foundation for intangible capital.** The paper anticipates later literatures on intangible assets, organizational capabilities, knowledge capital, firm routines, and the role of internal labor markets.
- **Policy implication.** Constraints on merger, takeover, and integration cannot be understood without understanding whether organization capital is transferable. This affects how one interprets antitrust and firm-boundary questions.

3 Model objects and measurement

3.1 Organization capital

Organization capital is the stock of information within a firm that affects production possibilities. It is capital because it is costly to acquire, persists over time, and raises future productivity. It is organizational because much of it is embedded in firm-specific relationships, routines, task assignments, and internal knowledge.

Interpretation: The paper treats information as an asset in the same conceptual class as physical or human capital. The key difference is that it is produced jointly with current operation and is difficult to transfer between firms.

3.2 Personnel information

- Workers have heterogeneous aptitudes that are not perfectly observed.
- Some tasks require specific worker types, so productivity depends on the match between worker and task.
- The firm learns about a worker's type by observing performance in a screening task.
- More observations imply a more precise posterior belief about worker type.

Interpretation: Personnel information turns the firm into a learning institution. The firm accumulates knowledge about where workers should be assigned.

3.3 Task structure

The simplified model uses three tasks. Task 1 is a screening or apprenticeship task; tasks 2 and 3 are production tasks in which costs depend on the worker's unobserved type. New workers start in task 1, and older vintage workers are eventually assigned to tasks 2 or 3.

Interpretation: The assignment structure creates the link between growth and costs. When a firm grows rapidly, it hires many new workers and cannot observe them long before placing them in productivity-sensitive tasks.

3.4 Signals and Bayesian learning

For worker i in period t , the firm observes a signal

$$z_{it} = \theta_i + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, 1).$$

After n observations, the posterior over the worker's type is summarized by a posterior mean m and precision h :

$$h = \pi + n, \quad m \propto \sum_{k=1}^n z_k.$$

Interpretation: More time in the screening task improves precision. Rapid growth shortens screening time and increases assignment mistakes.

3.5 Growth rate and screening time

Let g denote the output growth rate and p the quit rate. Faster growth raises the share of new and low-seniority employees. The time $n(g)$ that workers spend in the screening task falls as g rises.

$$\text{Higher } g \Rightarrow \text{lower } n(g) \Rightarrow \text{less information per worker.}$$

Interpretation: This is the core source of endogenous adjustment costs: the cost of growth is not a direct installation cost but a loss of information quality.

3.6 Team information

A second example treats workers as teams. Team productivity has an unobserved component. Observing team performance produces information about the quality of the worker pairing. Dissolving a team destroys the team-specific information.

Interpretation: Team information has the same structure as personnel information: faster expansion means less time to learn which pairings work.

3.7 Firm-specific human capital

The third example treats employee skills as organization capital. Current production and future human-capital accumulation are joint products. If the transformation frontier between current output and future human capital is concave, rapid accumulation is costly.

Interpretation: Human capital is not only an attribute of workers; some of it is embedded in firm-specific routines, practices, and relationships.

4 Models and theoretical specifications

4.1 Cost function in the personnel-information model

After n periods in the screening job, average cost falls with n :

$$c(n) = \bar{c} - \Delta(n), \quad \Delta'(n) > 0, \quad \Delta''(n) < 0.$$

The paper's normal-learning example yields a cost function that decreases at a decreasing rate with the number of observations.

Interpretation: More observations improve task assignment, but the marginal value of an additional observation declines. Therefore waiting longer before assigning workers is valuable, but not infinitely valuable.

4.2 Growth raises per-unit costs

The growth rate determines the vintage distribution of workers. Faster growth mechanically increases the share of workers with few observations. Hence:

$$g \uparrow \Rightarrow n(g) \downarrow \Rightarrow c(n(g)) \uparrow.$$

Interpretation: This is the precise logic behind endogenous adjustment costs. The adjustment cost is not imposed from outside; it is a consequence of the firm’s information-accumulation technology.

4.3 Technology constraints and constant returns

Let k_i be the beginning-of-period stock of employees with i observations, a_{ij} the number assigned to task j , and b the number of new hires. The technology constraints have the form:

$$a_{ij} = \phi_{ij}q, \quad \sum_j a_{ij} = k_i, \quad a_{ij} \geq 0,$$

with transition equations that update worker vintages after quits and screening.

Interpretation: The feasible set is a closed convex cone. This gives constant returns to scale while still preserving costly rapid adjustment through the vintage distribution of information.

4.4 Industry equilibrium and firm growth

The industry demand function is downward sloping. Firms are price takers in the model, and the equilibrium maximizes discounted consumer surplus subject to the organization-capital technology. Because the technology is homogeneous, the optimal growth rate is independent of firm size.

Interpretation: Firm size does not pin down the growth rate. What matters is the growth rate’s effect on information dilution within the firm.

4.5 Transferability and firm specificity

If organization capital could be transferred costlessly across firms, firms could merge, split, or pirate personnel without losing information. Then the model would not restrict growth. Therefore organization capital must be at least partly firm specific.

Interpretation: The transfer-cost section is essential. It explains why organization capital creates firm boundaries, not merely economy-wide knowledge.

5 Identification and theoretical design logic

This is a theoretical paper, so there is no econometric identification strategy. The design logic is instead conceptual and constructive: show that standard empirical regularities of firm growth can be generated by an information-accumulation technology without assuming adjustment costs directly.

5.1 What the paper explains

- Firms grow gradually rather than instantaneously.
- Growth rates can be approximately independent of firm size.
- Large firms can have lower variance of growth rates and profitability than small firms.
- Firm-specific information can explain why firm boundaries matter.
- Seniority rules and internal labor markets can have informational value.

5.2 Main assumptions that matter

- Worker and task characteristics are initially unknown.
- Information arrives through internal observation or experience.
- Rapid growth reduces the average precision of information used in assignments.
- Organization capital is at least partly firm specific and costly to transfer.
- Production has constant returns once organization capital is scaled proportionally.

5.3 Limitations

- The model abstracts from financing frictions, strategic market power, and managerial span of control.
- Labor turnover is suppressed in the main examples, even though turnover is important empirically.
- The model is not estimated or calibrated.
- Several mathematical structures are simplified to make proportional growth tractable.

Interpretation: These limitations make the model stylized, but they also isolate the paper’s key mechanism: information accumulation within firms can itself generate adjustment costs.

6 Section-by-section study guide

6.1 Introduction

The introduction defines organization capital and states the main claim: the firm is a storehouse of information, and the costs of building this information explain gradual growth and firm-size regularities.

6.2 Section I: Personnel Information as Organization Capital

This section develops the main example. Firms learn worker types through a screening task. Faster growth reduces screening time and raises assignment costs.

6.3 Section II: Team Information

This section shows that the same logic applies to team-specific knowledge. Team productivity is learned over time, and dissolving teams destroys accumulated information.

6.4 Section III: Firm-specific Human Capital

This section treats employee human capital as organization capital. Current output and future skill accumulation are joint products, and a concave transformation frontier generates costly rapid adjustment.

6.5 Section IV: Costs of Transferring Organization Capital

This section explains why organization capital must be firm specific. If it were costlessly transferable, mergers, divestitures, and poaching would eliminate growth constraints.

6.6 Section V: Summary and Concluding Remarks

The conclusion stresses that the firm facilitates information transmission and that understanding which information is processed inside firms can deepen the theory of firm behavior.

7 Tables, figures, and theoretical result blocks

The paper has no main empirical figures or tables. The closest analogues to tables/figures are the key model blocks and equations. The following blocks therefore summarize the paper’s main theoretical results.

7.1 Result Block 1: Organization Capital as Information

Definition Organization capital is firm-held information that affects the production possibility set and is produced jointly with output. It includes personnel information, team information, task information, and firm-specific human capital.

Read:

- Information is an asset because it improves production decisions in the future.
- It is organization capital because it resides in the firm, not only in individuals.
- It is costly to accumulate because learning uses current labor time and operational experience.

Detailed interpretation: This block is the central conceptual move of the paper. The authors convert “information in the firm” into a capital stock that can be accumulated, depreciated, and transferred only at a cost.

Economic message: The firm exists partly because it can produce and preserve information more effectively than arm’s-length market relations.

7.2 Result Block 2: Personnel Information and Screening

Learning technology A worker’s task aptitude is unobserved. A screening assignment produces signals $z_{it} = \theta_i + \varepsilon_{it}$. More observations raise posterior precision and improve assignment.

Read:

- Workers begin in a screening task.
- Senior workers have more observations and are assigned to tasks where mismatch is costly.
- Faster firm growth forces earlier assignment and less precise information.

Detailed interpretation: The screening model explains why internal labor markets and seniority have productive value: the firm knows more about older-vintage workers and can use them more efficiently.

Economic message: Rapid growth is costly because it dilutes the informational quality of the workforce.

7.3 Result Block 3: Endogenous Adjustment Costs

Growth cost Faster growth lowers the average screening time $n(g)$. Since per-unit costs $c(n)$ decline with n , faster growth raises per-unit costs.

Read:

- Adjustment costs are not assumed.
- They arise because growth changes the distribution of worker vintages and information precision.
- The cost of rapid expansion is an organizational learning cost.

Detailed interpretation: This is the paper's main technical contribution. It provides a microfoundation for convex adjustment costs in firm growth models.

Economic message: Firms grow gradually because information capital must be built inside the organization.

7.4 Result Block 4: Constant Returns and Gibrat's Law

Growth-size independence The feasible set is a convex cone, so the technology has constant returns to scale. With internal adjustment costs, the optimal growth rate can be independent of firm size.

Read:

- Constant returns explain why there is no unique optimum firm size.
- Internal adjustment costs explain why growth is not instantaneous.
- Together, these deliver a Gibrat-style relation between firm size and growth.

Detailed interpretation: The paper reconciles two facts that might seem contradictory: no strong scale economies at the firm level and yet costly adjustment in firm size.

Economic message: Scale can be replicated, but information cannot be instantly replicated.

7.5 Result Block 5: Team Information

Team-specific knowledge If team productivity is learned through observation and destroyed when teams dissolve, team information has the same mathematical structure as personnel information.

Read:

- Teams have match quality.
- Team signals accumulate through joint work.
- Dissolving teams destroys team-specific capital.

Detailed interpretation: The team example shows that the mechanism is not limited to individual worker assignment. Organization capital can exist in relationships among workers.

Economic message: Firms are repositories of relationship-specific knowledge, not just collections of individual skills.

7.6 Result Block 6: Firm-specific Human Capital

Joint production Current output and future human capital are joint products. With a concave transformation frontier between the two, rapid skill accumulation is costly.

Read:

- Learning and production occur together.
- Balanced mixtures of production and learning are more efficient than extreme specialization.
- Constant returns plus concavity again generate proportional growth.

Detailed interpretation: This block connects organization capital to human capital theory. The firm does not simply hire human capital; it produces firm-specific human capital through work.

Economic message: The firm's growth constraint comes from the time and interaction needed to build skills and routines.

7.7 Result Block 7: Transfer Costs and Firm Boundaries

Firm specificity If organization capital were costlessly transferable, the model would not constrain growth. Organization capital must therefore be partly firm specific.

Read:

- Moving people, teams, routines, and relationships between firms is costly.
- Organization capital includes tacit knowledge such as who to ask, how internal systems work, and how buyers and sellers interact.
- Transfer costs create firm boundaries and make internal growth different from acquisition or poaching.

Detailed interpretation: This is the bridge from adjustment costs to the theory of the firm. Organization capital matters for firm boundaries only if it cannot be freely moved.

Economic message: A firm is partly defined by the nontransferability of its internal information stock.

8 Detailed result map

- **Organization capital definition.** Information is a productive asset held by the firm.
- **Personnel information mechanism.** Firms learn worker types through internal observation.

- **Rapid-growth cost.** Faster growth means less information per worker before task assignment.
- **Constant returns.** The technology can be replicated proportionally once organization capital is scaled.
- **Gibrat’s Law.** A size-independent optimal growth rate emerges from constant returns plus internal adjustment costs.
- **Team information.** Worker-pairing information creates the same adjustment-cost logic.
- **Firm-specific human capital.** Skill accumulation through work generates a concave output-capital frontier.
- **Transfer costs.** Organization capital constrains growth only if it is difficult to transfer across firms.
- **Firm theory.** The firm is a structure for acquiring, storing, and transmitting information.

9 Short summary

- Prescott and Visscher introduce the concept of organization capital.
- Organization capital is information inside the firm that affects production possibilities.
- The paper studies personnel information, team information, and firm-specific human capital.
- In each case, current production must be traded off against future organizational information.
- Faster growth reduces the amount or quality of information available per worker/team/task.
- This creates endogenous costs of rapid adjustment.
- The model can generate a firm growth rate independent of firm size, consistent with Gibrat’s Law.
- Organization capital must be costly to transfer, otherwise mergers, divestitures, and poaching would remove the growth constraint.
- The paper’s central insight is that the firm is an information-processing institution.

10 Conclusion

10.1 Core conclusion

Prescott and Visscher argue that firms exist and grow gradually because they accumulate organization capital: firm-specific information about workers, teams, tasks, and human capital. This information is valuable for production and costly to build quickly.

10.2 Economic intuition

The simplest intuition is the apprenticeship example. A firm can learn who is good at what by observing workers in screening tasks. If it grows too fast, it must assign workers before enough is known about them. This raises production costs. The same logic applies to teams and firm-specific human capital.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to provide a microfoundation for adjustment costs and gradual firm growth from internal information accumulation. Instead of imposing convex adjustment costs, it shows how they arise from the technology of learning inside firms.

10.4 Implications for the theory of the firm

The paper suggests that the firm is not only a contractual arrangement or production unit. It is an institutional setting in which agents have incentives to acquire and reveal information at lower cost than in alternative market arrangements.

10.5 Implications for firm growth and size distributions

The model helps explain why firm growth rates can be approximately independent of size and why larger firms may exhibit less volatile growth and profitability. Large firms act like portfolios of smaller subunits whose errors partially diversify.

10.6 Limitations

The paper is theoretical and stylized. It abstracts from financing frictions, market power, explicit managerial agency problems, and labor turnover. It does not estimate organization capital empirically. Its purpose is to isolate a mechanism, not to provide a complete quantitative model.

10.7 Takeaway

The enduring lesson is that organizational information is a form of capital. Firms grow slowly not just because machines or funds are costly, but because knowledge of people, teams, tasks, and routines must be accumulated inside the organization over time.